

Probability & Random Processes EE2011

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Expectation, Variances

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4.1 Mean of a Random Variable

consider the following.

If two coins are tossed 16 times and X is the number of heads that occur per toss, then the values of X are 0, 1, and 2.

Suppose that the experiment yields no heads, one head, and two heads a total of 4, 7, and 5 times, respectively.

The average number of heads per toss of the two coins is then

$$\frac{(0)(4) + (1)(7) + (2)(5)}{16} = 1.06.$$

This is an average value of the data and yet it is not a possible outcome of $\{0, 1, 2\}$.

Hence, an average is not necessarily a possible outcome for the experiment.



Let us now restructure our computation for the average number of heads so as to have the following equivalent form:

$$(0) \left(\frac{4}{16} \right) + (1) \left(\frac{7}{16} \right) + (2) \left(\frac{5}{16} \right) = 1.06.$$

The numbers $4/16$, $7/16$, and $5/16$ are the fractions of the total tosses resulting in 0, 1, and 2 heads, respectively.

These fractions are also the relative frequencies for the different values of X in our experiment.

In fact, then, we can calculate the mean, or average, of a set of data by knowing the distinct values that occur and their relative frequencies, without any knowledge of the total number of observations in our set of data.

Therefore, if $4/16$, or $1/4$, of the tosses result in no heads, $7/16$ of the tosses result in one head, and $5/16$ of the tosses result in two heads, the mean number of heads per toss would be 1.06 no matter whether the total number of tosses were 16, 1000, or even 10,000.



This method of relative frequencies is used to calculate the average number of heads per toss of two coins that we might expect in the long run.

We shall refer to this average value as the **mean of the random variable X** or the **mean of the probability distribution of X** and write it as μ_x or simply as μ when it is clear to which random variable we refer.

It is also common among statisticians to refer to this mean as the mathematical expectation, or the expected value of the random variable X , and denote it as $E(X)$.



Assuming that 1 fair coin was tossed twice, we find that the sample space for our experiment is

$$S = \{HH, HT, TH, TT\}.$$

Since the 4 sample points are all equally likely, it follows that

$$P(X = 0) = P(TT) = \frac{1}{4}, \quad P(X = 1) = P(TH) + P(HT) = \frac{1}{2},$$

and
$$P(X = 2) = P(HH) = \frac{1}{4},$$

where a typical element, say TH , indicates that the first toss resulted in a tail followed by a head on the second toss.

Now, these probabilities are just the relative frequencies for the given events in the long run.

Therefore,
$$\mu = E(X) = (0) \left(\frac{1}{4}\right) + (1) \left(\frac{1}{2}\right) + (2) \left(\frac{1}{4}\right) = 1.$$

This result means that a person who tosses 2 coins over and over again will, on the average, get 1 head per toss.



The method described above for calculating the expected number of heads per toss of 2 coins suggests that

the mean, or expected value, of any discrete random variable may be obtained by multiplying each of the values $x_1, x_2, \dots, x_n$ of the random variable X by its corresponding probability $f(x_1), f(x_2), \dots, f(x_n)$ and summing the products.

This is true, however, only if the random variable is discrete.

In the case of continuous random variables, the definition of an expected value is essentially the same with summations replaced by integrations.



Definition 4.1:

Let X be a random variable with probability distribution $f(x)$. The **mean**, or **expected value**, of X is

$$\mu = E(X) = \sum_x x f(x)$$

if X is discrete, and

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

if X is continuous.

the mean is usually understood as a “center” value of the underlying distribution if we use the expected value, as in Definition 4.1.



Example 4.1:

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components.

A sample of 3 is taken by the inspector.

Find the expected value of the number of good components in this sample.

Solution:

Let X represent the number of good components in the sample.

The probability distribution of X is

$$f(x) = \frac{\binom{4}{x} \binom{3}{3-x}}{\binom{7}{3}}, \quad x = 0, 1, 2, 3.$$

Simple calculations yield $f(0) = 1/35$, $f(1) = 12/35$, $f(2) = 18/35$, and $f(3) = 4/35$. Therefore,

$$\mu = E(X) = (0) \left(\frac{1}{35} \right) + (1) \left(\frac{12}{35} \right) + (2) \left(\frac{18}{35} \right) + (3) \left(\frac{4}{35} \right) = \frac{12}{7} = 1.7.$$

Thus, if a sample of size 3 is selected at random over and over again from a lot of 4 good components and 3 defective components, it will contain, on average, 1.7 good components.



Example 4.2:

A salesperson for a medical device company has two appointments on a given day.

At the first appointment, he believes that he has a 70% chance to make the deal, from which he can earn \$1000 commission if successful.

On the other hand, he thinks he only has a 40% chance to make the deal at the second appointment, from which, if successful, he can make \$1500.

What is his expected commission based on his own probability belief?

Assume that the appointment results are independent of each other.

Solution:

First, we know that the salesperson, for the two appointments, can have 4 possible commission totals: \$0, \$1000, \$1500, and \$2500.

We then need to calculate their associated probabilities.



By independence, we obtain

$$\begin{aligned}f(\$0) &= (1 - 0.7)(1 - 0.4) = 0.18, & f(\$2500) &= (0.7)(0.4) = 0.28, \\f(\$1000) &= (0.7)(1 - 0.4) = 0.42, & \text{and } f(\$1500) &= (1 - 0.7)(0.4) = 0.12.\end{aligned}$$

Therefore, the expected commission for the salesperson is

$$\begin{aligned}E(X) &= (\$0)(0.18) + (\$1000)(0.42) + (\$1500)(0.12) + (\$2500)(0.28) \\&= \$1300.\end{aligned}$$



Example 4.3:

Let X be the random variable that denotes the life in hours of a certain electronic device.

The probability density function is

$$f(x) = \begin{cases} \frac{20,000}{x^3}, & x > 100, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected life of this type of device.

Solution:

Using Definition 4.1, we have

$$\mu = E(X) = \int_{100}^{\infty} x \frac{20,000}{x^3} dx = \int_{100}^{\infty} \frac{20,000}{x^2} dx = 200.$$

Therefore, we can expect this type of device to last, *on average*, 200 hours. └



Now let us consider a new random variable $g(X)$, which depends on X ; that is, each value of $g(X)$ is determined by the value of X .

For instance, $g(X)$ might be X^2 or $3X - 1$, and whenever X assumes the value 2, $g(X)$ assumes the value $g(2)$.

In particular, if X is a discrete random variable with probability distribution $f(x)$, for $x = -1, 0, 1, 2$, and $g(X) = X^2$, then

$$\begin{array}{cccc}
 x = & -1, & 0, & 1, & 2, \\
 f(x) = & f(-1) & f(0) & f(1) & f(2) \\
 g(X) = X^2 = & 1 & 0 & 1 & 4 \\
 g(x) = & 0 & 1 & & 4 \\
 f(g(X) = g(x)) = & f(0) & f(-1)+f(1) & & f(2)
 \end{array}$$

the probability distribution of $g(X)$ may be written

$g(x)$	0	1	4
$P[g(X) = g(x)]$	$f(0)$	$f(-1) + f(1)$	$f(2)$



By the definition of the expected value of a random variable,

$$\mu = E(X) = \sum_x x f(x)$$

we obtain

$$\begin{aligned}\mu_{g(X)} &= E[g(x)] : \\ &= 0f(0) + 1[f(-1) + f(1)] + 4f(2) \\ &= (-1)^2 f(-1) + (0)^2 f(0) + (1)^2 f(1) + (2)^2 f(2) \\ &= \sum g(x) f(x).\end{aligned}$$

This result is generalized in Theorem 4.1 for both discrete and continuous random variables.



if X is a discrete random variable with probability distribution $f(x)$, for $x = 0, 1, 2$, and $g(X) = 3X - 1$, then

$$P[g(X) = -1] = P(X = 0) = f(0),$$

$$P[g(X) = 2] = P(X = 1) = f(1),$$

$$P[g(X) = 5] = P(X = 2) = f(2),$$



Theorem 4.1:

Let X be a random variable with probability distribution $f(x)$. The expected value of the random variable $g(X)$ is

$$\mu_{g(X)} = E[g(X)] = \sum_x g(x)f(x)$$

if X is discrete, and

$$\mu_{g(X)} = E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx$$

if X is continuous.



Example 4.4:

Suppose that the number of cars X that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

x	4	5	6	7	8	9
$P(X = x)$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Let $g(X) = 2X - 1$ represent the amount of money, in dollars, paid to the attendant by the manager.

Find the attendant's expected earnings for this particular time period.

Solution:

By Theorem 4.1, the attendant can expect to receive

$$\begin{aligned} E[g(X)] &= E(2X - 1) = \sum_{x=4}^9 (2x - 1)f(x) \\ &= (7) \left(\frac{1}{12} \right) + (9) \left(\frac{1}{12} \right) + (11) \left(\frac{1}{4} \right) + (13) \left(\frac{1}{4} \right) \\ &\quad + (15) \left(\frac{1}{6} \right) + (17) \left(\frac{1}{6} \right) = \$12.67. \end{aligned}$$



Example 4.5: Let X be a random variable with density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of $g(X) = 4X + 3$.

Solution:

By Theorem 4.1, we have

$$\begin{aligned} E(4X + 3) &= \int_{-1}^2 \frac{(4x + 3)x^2}{3} dx \\ &= \frac{1}{3} \int_{-1}^2 (4x^3 + 3x^2) dx = 8. \end{aligned}$$



We shall now extend our concept of mathematical expectation to the case of two random variables X and Y with joint probability distribution $f(x, y)$.

If $g(X, Y)$ is a function of these two random variables, then its expected value is given by the following:

Definition 4.2:

Let X and Y be random variables with joint probability distribution $f(x, y)$. The mean, or expected value, of the random variable $g(X, Y)$ is

$$\mu_{g(X, Y)} = E[g(X, Y)] = \sum_x \sum_y g(x, y) f(x, y)$$

if X and Y are discrete, and

$$\mu_{g(X, Y)} = E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy$$

if X and Y are continuous.

We now look at taking the expectation of jointly distributed discrete random variables. Because expected values are defined for a single quantity, we will actually define the expected value of a combination of the pair of random variables, i.e., we look at the expected value of a function applied to (X, Y)

Generalization of Definition 4.2 for the calculation of mathematical expectations of functions of several random variables is straightforward.



Example 4.6: Let X and Y be the random variables with joint probability distribution indicated in Table 3.1 on page 96. Find the expected value of $g(X, Y) = XY$. The table is reprinted here for convenience.

$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Solution: By Definition 4.2,

$$\mu_{g(X,Y)} = E[g(X, Y)] = \sum_x \sum_y g(x, y) f(x, y)$$

we write

$$\begin{aligned}
 E(XY) &= \sum_{x=0}^2 \sum_{y=0}^2 xy f(x, y) \\
 &= (0)(0)f(0, 0) + (0)(1)f(0, 1) \\
 &\quad + (1)(0)f(1, 0) + (1)(1)f(1, 1) + (2)(0)f(2, 0) \\
 &= f(1, 1) = \frac{3}{14}.
 \end{aligned}$$



Example 4.7: Find $E(Y/X)$ for the density function

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Solution: We have

$$\mu_{g(X,Y)} = E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy$$

$$E\left(\frac{Y}{X}\right) = \int_0^1 \int_0^2 \frac{y}{x} \cdot \frac{x(1+3y^2)}{4} dx dy$$

$$= \int_0^1 \int_0^2 \frac{y(1+3y^2)}{4} dx dy = \int_0^1 \frac{y + 3y^3}{2} dy = \frac{5}{8}.$$



Note that if $g(X, Y) = X$ in Definition 4.2, we have

$$E(X) = \begin{cases} \sum_x \sum_y x f(x, y) = \sum_x x g(x) & \text{(discrete case),} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dy dx = \int_{-\infty}^{\infty} x g(x) dx & \text{(continuous case),} \end{cases}$$

where $g(x)$ is the marginal distribution of X .

Therefore, in calculating $E(X)$ over a two-dimensional space, one may use either the joint probability distribution of X and Y or the marginal distribution of X .

Similarly, we define

$$E(Y) = \begin{cases} \sum_y \sum_x y f(x, y) = \sum_y y h(y) & \text{(discrete case),} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy = \int_{-\infty}^{\infty} y h(y) dy & \text{(continuous case),} \end{cases}$$

where $h(y)$ is the marginal distribution of the random variable Y .



4.2 Variance and Covariance of Random Variables

The mean, or expected value, of a random variable X is of special importance in statistics because it describes where the probability distribution is centered.

By itself, however, the mean does not give an adequate description of the shape of the distribution.

We also need to characterize the variability in the distribution.



In Figure 4.1,

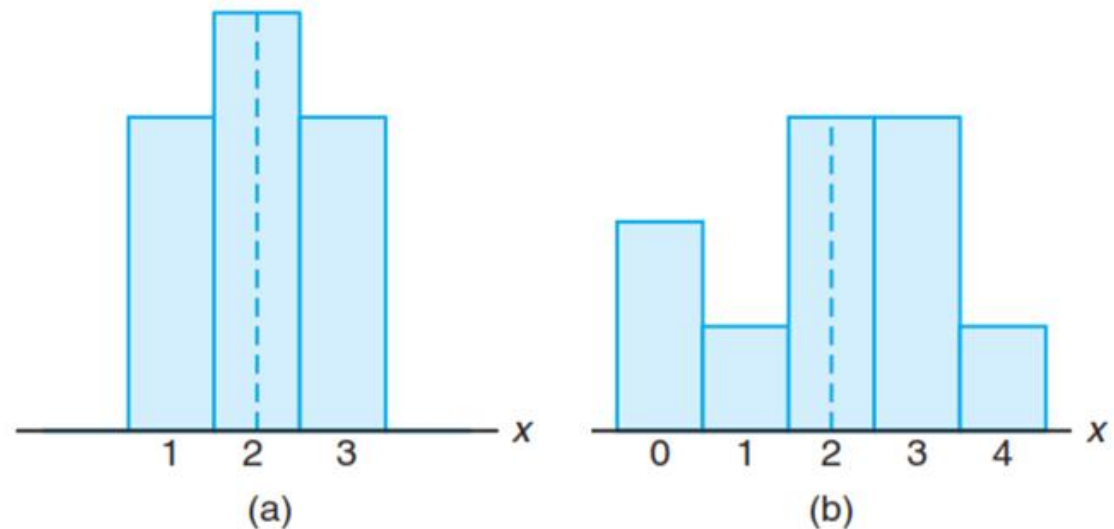


Figure 4.1: Distributions with equal means and unequal dispersions.

we have the histograms of two discrete probability distributions that have the same mean, $\mu = 2$, but differ considerably in variability, or the dispersion of their observations about the mean.

The most important measure of variability of a random variable X is obtained by applying Theorem 4.1 with $g(X) = (X - \mu)^2$.

The quantity is referred to as the **variance of the random variable X**

or the **variance of the probability distribution of X**

it is denoted by $\text{Var}(X)$ or the symbol σ_X^2 ,

or simply by σ^2 when it is clear to which random variable we refer.



Definition 4.3:

Let X be a random variable with probability distribution $f(x)$ and mean μ . The variance of X is

$$\sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x), \quad \text{if } X \text{ is discrete, and}$$

$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx, \quad \text{if } X \text{ is continuous.}$$

The positive square root of the variance, σ , is called the **standard deviation** of X .

The quantity $x - \mu$ in Definition 4.3 is called the **deviation of an observation** from its mean.

Since the deviations are squared and then averaged, σ^2 will be much smaller for a set of x values that are close to μ than it will be for a set of values that vary considerably from μ .



Example 4.8:

Let the random variable X represent the number of automobiles that are used for official business purposes on any given workday.

The probability distribution for company A [Figure 4.1(a)] is

x	1	2	3
$f(x)$	0.3	0.4	0.3

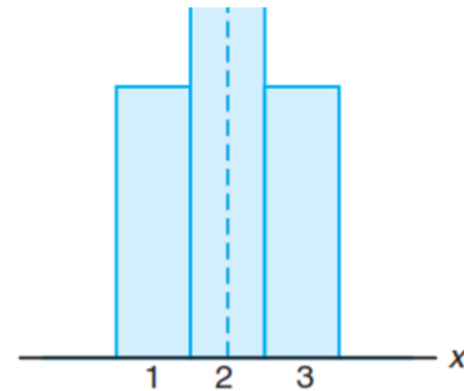


Figure 4.1: (a)

and that for company B [Figure 4.1(b)] is

x	0	1	2	3	4
$f(x)$	0.2	0.1	0.3	0.3	0.1

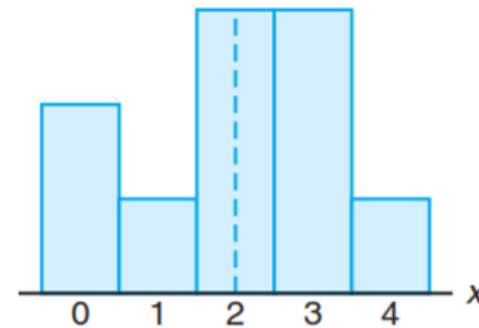


Figure 4.1: (b)

Show that the variance of the probability distribution for company B is greater than that for company A .

Solution: For company A , we find that

$$\mu_A = E(X) = (1)(0.3) + (2)(0.4) + (3)(0.3) = 2.0,$$

and then

$$\sigma_A^2 = \sum_{x=1}^3 (x-2)^2 f(x) = (1-2)^2(0.3) + (2-2)^2(0.4) + (3-2)^2(0.3) = 0.6.$$

For company B , we have

$$\mu_B = E(X) = (0)(0.2) + (1)(0.1) + (2)(0.3) + (3)(0.3) + (4)(0.1) = 2.0,$$

and then

$$\begin{aligned}\sigma_B^2 &= \sum_{x=0}^4 (x-2)^2 f(x) \\ &= (0-2)^2(0.2) + (1-2)^2(0.1) + (2-2)^2(0.3) \\ &\quad + (3-2)^2(0.3) + (4-2)^2(0.1) = 1.6.\end{aligned}$$

Clearly, the variance of the number of automobiles that are used for official business purposes is greater for company B than for company A .



An alternative and preferred formula for finding σ^2 , which often simplifies the calculations, is stated in the following theorem.

Theorem 4.2:

The variance of a random variable X is

$$\sigma^2 = E(X^2) - \mu^2.$$

Proof:

For the discrete case, we can write

$$\begin{aligned}\sigma^2 &= \sum_x (x - \mu)^2 f(x) = \sum_x (x^2 - 2\mu x + \mu^2) f(x) \\ &= \sum_x x^2 f(x) - 2\mu \sum_x x f(x) + \mu^2 \sum_x f(x).\end{aligned}$$

Since $\mu = \sum_x x f(x)$ by definition, and $\sum_x f(x) = 1$ for any discrete probability distribution,

it follows that
$$\sigma^2 = \sum_x x^2 f(x) - \mu^2 = E(X^2) - \mu^2.$$

For the continuous case the proof is step by step the same, with summations replaced by integrations.



Example 4.9: Let the random variable X represent the number of defective parts for a machine when 3 parts are sampled from a production line and tested.

The following is the probability distribution of X .

x	0	1	2	3
$f(x)$	0.51	0.38	0.10	0.01

Using Theorem 4.2, calculate σ^2 .

Solution: First, we compute

$$\mu = (0)(0.51) + (1)(0.38) + (2)(0.10) + (3)(0.01) = 0.61.$$

Now,

$$E(X^2) = (0)(0.51) + (1)(0.38) + (4)(0.10) + (9)(0.01) = 0.87.$$

Therefore,

$$\sigma^2 = 0.87 - (0.61)^2 = 0.4979.$$



Example 4.10: The weekly demand for a drinking-water product, in thousands of liters, from a local chain of efficiency stores is a continuous random variable X having the probability density

$$f(x) = \begin{cases} 2(x-1), & 1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the mean and variance of X .

Solution: Calculating $E(X)$ and $E(X^2)$, we have

$$\mu = E(X) = 2 \int_1^2 x(x-1) dx = \frac{5}{3}$$

and

$$E(X^2) = 2 \int_1^2 x^2(x-1) dx = \frac{17}{6}.$$

Therefore,

$$\sigma^2 = \frac{17}{6} - \left(\frac{5}{3}\right)^2 = \frac{1}{18}.$$



At this point, the variance or standard deviation has meaning only when we compare two or more distributions that have the same units of measurement.

Therefore, we could compare the variances of the distributions of contents, measured in liters, of bottles of orange juice from two companies, and the larger value would indicate the company whose product was more variable or less uniform.

It would not be meaningful to compare the variance of a distribution of heights to the variance of a distribution of aptitude scores.



We shall now extend our concept of the variance of a random variable X to include random variables related to X .

For the random variable $g(X)$, the variance is denoted by $\sigma_{g(X)}^2$ and is calculated by means of the following theorem.

Theorem 4.3:

Let X be a random variable with probability distribution $f(x)$. The variance of the random variable $g(X)$ is

$$\sigma_{g(X)}^2 = E\{[g(X) - \mu_{g(X)}]^2\} = \sum_x [g(x) - \mu_{g(X)}]^2 f(x)$$

if X is discrete, and

$$\sigma_{g(X)}^2 = E\{[g(X) - \mu_{g(X)}]^2\} = \int_{-\infty}^{\infty} [g(x) - \mu_{g(X)}]^2 f(x) dx$$

if X is continuous.



Proof:

Since $g(X)$ is itself a random variable with mean $\mu_{g(X)}$ as defined in Theorem 4.1,

it follows from Definition 4.3 that

$$\sigma_{g(X)}^2 = E\{[g(X) - \mu_{g(X)}]^2\}.$$

Now, applying Theorem 4.1 again to the random variable $[g(X) - \mu_{g(X)}]^2$ completes the proof. 



Example 4.11: Calculate the variance of $g(X) = 2X + 3$, where X is a random variable with probability distribution

x	0	1	2	3
$f(x)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{8}$

Solution: First, we find the mean of the random variable $2X + 3$.

According to Theorem 4.1,

$$\mu_{2X+3} = E(2X + 3) = \sum_{x=0}^3 (2x + 3)f(x) = 6.$$

Now, using Theorem 4.3, we have

$$\begin{aligned}\sigma_{2X+3}^2 &= E\{[(2X + 3) - \mu_{2X+3}]^2\} = E[(2X + 3 - 6)^2] \\ &= E(4X^2 - 12X + 9) = \sum_{x=0}^3 (4x^2 - 12x + 9)f(x) = 4.\end{aligned}$$



Example 4.12: Let X be a random variable having the density function given in Example 4.5 on page 115.

Find the variance of the random variable $g(X) = 4X + 3$.

Solution: In Example 4.5, we found that $\mu_{4X+3} = 8$.

Now, using Theorem 4.3,

$$\sigma_{4X+3}^2 = E\{[(4X + 3) - 8]^2\} = E[(4X - 5)^2]$$

$$= \int_{-1}^2 (4x - 5)^2 \frac{x^2}{3} dx$$

$$= \frac{1}{3} \int_{-1}^2 (16x^4 - 40x^3 + 25x^2) dx = \frac{51}{5}.$$



If $g(X, Y) = (X - \mu_X)(Y - \mu_Y)$, where $\mu_X = E(X)$ and $\mu_Y = E(Y)$, Definition 4.2 yields an expected value called the **covariance** of X and Y , which we denote by σ_{XY} or $\text{Cov}(X, Y)$.

Definition 4.4:

Let X and Y be random variables with joint probability distribution $f(x, y)$. The covariance of X and Y is

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f(x, y)$$

if X and Y are discrete, and

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y) dx dy$$

if X and Y are continuous.



The covariance between two random variables is a measure of the nature of the association between the two.

If large values of X often result in large values of Y or small values of X result in small values of Y , positive $X - \mu_X$ will often result in positive $Y - \mu_Y$ and negative $X - \mu_X$ will often result in negative $Y - \mu_Y$.

Thus, the product $(X - \mu_X)(Y - \mu_Y)$ will tend to be positive.

On the other hand, if large X values often result in small Y values, the product $(X - \mu_X)(Y - \mu_Y)$ will tend to be negative.

The *sign* of the covariance indicates whether the relationship between two dependent random variables is positive or negative.



When X and Y are statistically independent, it can be shown that the covariance is zero (see Corollary 4.5).

The converse, however, is not generally true.

Two variables may have zero covariance and still not be statistically independent.

Note that the covariance only describes the *linear* relationship between two random variables.

Therefore, if a covariance between X and Y is zero, X and Y may have a nonlinear relationship, which means that they are not necessarily independent.



The alternative and preferred formula for σ_{XY} is stated by Theorem 4.4.

Theorem 4.4:

The covariance of two random variables X and Y with means μ_X and μ_Y , respectively, is given by

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y.$$

Proof:

For the discrete case, we can write

$$\begin{aligned}\sigma_{XY} &= \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f(x, y) \\ &= \sum_x \sum_y xy f(x, y) - \mu_X \sum_x \sum_y y f(x, y) \\ &\quad - \mu_Y \sum_x \sum_y x f(x, y) + \mu_X \mu_Y \sum_x \sum_y f(x, y).\end{aligned}$$



Since

$$\mu_X = \sum_x xf(x, y), \quad \mu_Y = \sum_y yf(x, y), \quad \text{and} \quad \sum_x \sum_y f(x, y) = 1$$

for any joint discrete distribution,

it follows that

$$\sigma_{XY} = E(XY) - \mu_X\mu_Y - \mu_Y\mu_X + \mu_X\mu_Y = E(XY) - \mu_X\mu_Y.$$

For the continuous case, the proof is identical with summations replaced by integrals. └



Example 4.13:

Example 3.14 on page 95 describes a situation involving the number of blue refills X and the number of red refills Y .

Two refills for a ballpoint pen are selected at random from a certain box, and the following is the joint probability distribution:

$f(x, y)$		x			$h(y)$
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
$g(x)$		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Find the covariance of X and Y .



Solution:

From Example 4.6, we see that $E(XY) = 3/14$.

Now

$$\mu_X = \sum_{x=0}^2 xg(x) = (0) \left(\frac{5}{14}\right) + (1) \left(\frac{15}{28}\right) + (2) \left(\frac{3}{28}\right) = \frac{3}{4},$$

and

$$\mu_Y = \sum_{y=0}^2 yh(y) = (0) \left(\frac{15}{28}\right) + (1) \left(\frac{3}{7}\right) + (2) \left(\frac{1}{28}\right) = \frac{1}{2}.$$

Therefore,

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y = \frac{3}{14} - \left(\frac{3}{4}\right) \left(\frac{1}{2}\right) = -\frac{9}{56}.$$



Example 4.14:

The fraction X of male runners and the fraction Y of female runners who compete in marathon races are described by the joint density function

$$f(x, y) = \begin{cases} 8xy, & 0 \leq y \leq x \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the covariance of X and Y .

Solution:

We first compute the marginal density functions.

They are

$$g(x) = \begin{cases} 4x^3, & 0 \leq x \leq 1, \\ 0, & \text{elsewhere,} \end{cases} \quad \text{and} \quad h(y) = \begin{cases} 4y(1 - y^2), & 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

From these marginal density functions, we compute

$$\mu_X = E(X) = \int_0^1 4x^4 \, dx = \frac{4}{5} \quad \text{and} \quad \mu_Y = \int_0^1 4y^2(1 - y^2) \, dy = \frac{8}{15}.$$

From the joint density function given above, we have

$$E(XY) = \int_0^1 \int_y^1 8x^2y^2 \, dx \, dy = \frac{4}{9}.$$

Then

$$\sigma_{XY} = E(XY) - \mu_X\mu_Y = \frac{4}{9} - \left(\frac{4}{5}\right)\left(\frac{8}{15}\right) = \frac{4}{225}.$$

